Lab 02 - Regression

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1. Introduction

This project is based on the Kaggle competition *"mlX 2.0 Regression Challenge: Predict the Hype"* where the objective is to predict a song's popularity score (0–100) using audio features and metadata. The dataset includes features such as energy, danceability, tempo, and other derived metrics. Two separate submission pipelines were developed for this task.

- Submission 1 used a baseline Linear Regression model alongside stronger boosting-based models such as CatBoost, XGBoost, LightGBM, and Gradient Boosting with blending.
- Submission 2 used a focused LightGBM approach with frequency encoding, target encoding, and additional engineered features.

RMSE was the main competition metric, but MAE, R2, and MAPE were also computed to give a broader view of model performance.

2. Dataset

- The dataset consists of three files: train.csv, test.csv, sample_submission.csv
- Each row represents a music track with:
    - Audio features (energy, tempo, danceability)
    - Derived features (emotional charge, groove efficiency)
    - Metadata (release date, artist details)
- **Dataset Details:** training samples: 61,609, test samples: 41,074, features: 61
- **Target Variable:** Continuous popularity score (0–100)

Additional details: https://www.kaggle.com/competitions/mlx-2-0-regression/data

3. Data preprocessing

Submission 1

Submission 1 performs feature engineering directly on the dataframe before modeling. The publication_timestamp is converted to datetime and expanded into year, month, day, and weekday features. Additional aggregated and interaction features are created from related feature groups. Identifier and label columns such as track_identifier and composition_label_0/1/2 are removed.

For preprocessing, numeric values are median-imputed, categorical values are filled with "Unknown", and categorical variables are one-hot encoded using a ColumnTransformer.

Submission 2

Submission 2 uses a more compact and advanced preprocessing approach. It removes track_identifier, converts publication_timestamp into datetime, and extracts year and month. The composition_label_0/1/2 columns are frequency encoded.

Missing values are handled using median imputation (numeric) and "Unknown" (categorical). Categorical features are encoded based on cardinality using ordinal encoding (low-cardinality) and target encoding (high-cardinality). Additional aggregated and interaction features are also included before training a LightGBM model.

4. Models Used

Model	Description	Why Chosen	Advantages	Disadvantages
Linear Regression (Submission 1 Baseline)	A baseline regression model that predicts popularity as a weighted linear combination of input features after preprocessing.	Used as a fast and interpretable benchmark to compare against more complex models.	Very fast training and inference, simple to interpret, low computational cost.	Cannot capture non-linear relationships or complex feature interactions, leading to lower accuracy on rich datasets.

Advanced Boosting Models (Submission 1: CatBoost, XGBoost, LightGBM, Gradient Boosting + Blending)	Ensemble models that build boosted decision trees; multiple models are compared and optionally blended for better performance.	Chosen to capture non-linear patterns and feature interactions and improve leaderboard performance.	High predictive power, handles tabular data well, strong performance in competitions.	More complex to tune, higher training time, less interpretable, risk of overfitting.
LightGBM Focused Pipeline (Submission 2)	A single optimized LightGBM model trained on engineered and encoded features (frequency, ordinal, target encoding).	Selected for a strong balance between accuracy and efficiency on tabular data.	Fast training, scalable, high accuracy with proper feature engineering.	Sensitive to hyperparameters and preprocessing, less interpretable than simple models.

5. Evaluation Metrics

1. Root Mean Squared Error (RMSE): Main competition metric, Penalizes large errors
2. Mean Absolute Error (MAE): Average absolute difference between predictions and actual values
3. R^2 Score: Measures variance explained by the model
4. Mean Absolute Percentage Error (MAPE): Measures relative error percentage

6. Results

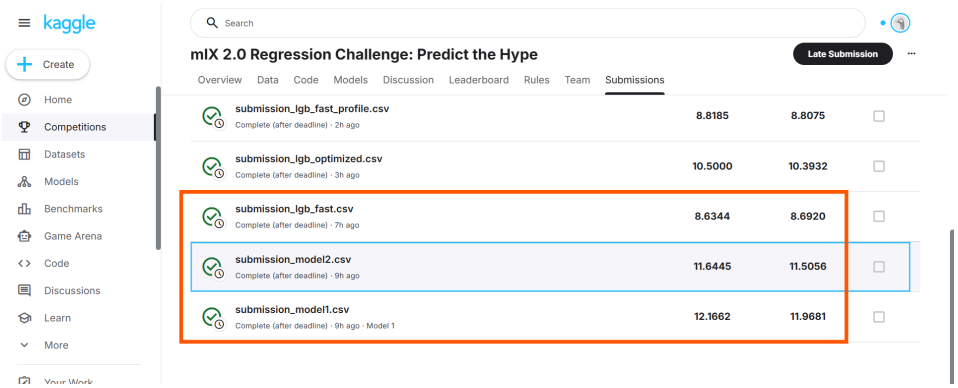
Validation Performance

Submission	Model	RMSE	MAE	R^2	Notes
Submission 1	Linear Regression	12.41	8.18	0.67	Baseline model
Submission 1	CatBoost	11.75	8.55	0.70	Improved over baseline
Submission 1	Blended Model	11.49	–	–	Best validation result in Submission 1
Submission 2	LightGBM	10.08	5.43	0.78	Best overall validation performance

Kaggle Leaderboard Results

Submission	File Name	Private Score (RMSE)	Public Score (RMSE)
Submission 1	submission_model1.csv	12.17	11.97
Submission 1	submission_model2.csv	11.64	11.51
Submission 2	submission_lgb_fast.csv	8.63	8.69

7. Conclusion

Submission 2 clearly achieved the best performance, with substantially lower RMSE on both validation and leaderboard datasets. This improvement is mainly due to the use of **LightGBM combined with more effective preprocessing and encoding strategies**, which better capture non-linear relationships in the data.

8. Screenshot of the submission